BT-205 (GS)

B.Tech., I & II Semester Examination, June 2026

Grading System (GS)

BASIC COMPUTER ENGINEERING

Time : Three Hours Maximum Marks : 70

Note : (i) Attempt any five questions. (ii) All questions carry equal marks. (iii) Original paper bilingual (Hindi + English) tha; yahan sirf English version diya gaya hai.

Coverage: Full paper (Q1–Q8)

1. (a) What is an Operating System? Discuss the major functions performed by an operating system. Explain in detail how it manages files, processes and memory in a computer system. **7**

(b) Explain the main features of MS-Word, MS-Excel and MS-PowerPoint. Discuss their importance in office automation. **7**

2. (a) Explain the role of computers in any four of the following areas with suitable examples: E-Business, Bioinformatics, Healthcare, Remote Sensing & GIS, Meteorology, Multimedia and Animation. **7**

(b) Discuss the fundamental characteristics of Object Oriented Programming. How does Object Oriented Programming overcome the limitations of Procedure Oriented Programming? **7**

3. (a) Define an algorithm. Explain the characteristics of a good algorithm. Draw a flowchart to find the largest of three numbers and discuss the advantages of flowcharts in program development. **7**

(b) What is an array in C++? Explain different types of functions in C++ with suitable examples. **7**

4. (a) What is the scope resolution operator? Explain its applications in C++ with suitable examples. Distinguish between local and global variables. **7**

(b) Explain inheritance in C++. Discuss different types of inheritance with suitable diagrams and examples. **7**

5. (a) Explain the concept of polymorphism in C++. Discuss function overloading, operator overloading and virtual functions with suitable examples. **7**

(b) Explain the objectives of computer networking. Describe the ISO-OSI reference model with the functions of each layer. **7**

6. (a) What is computer security? Explain the threats of viruses, worms, Trojans and malware. **7**

(b) Explain the role of Firewall in computer security. Discuss computer ethics and cyber laws. **7**

7. (a) Explain the role and responsibilities of a Database Administrator (DBA). Discuss the concept of primary key with suitable examples. **7**

(b) What is Cloud Computing? Explain the cloud infrastructure and service delivery models (IaaS, PaaS and SaaS). **7**

8. Write a short note on any two: **14**

(a) Memory Hierarchy (b) Friend Function (c) World Wide Web (WWW) (d) Data Dictionary

****

Yeh reconstructed paper kya hai? RGPV ke 2022–2026 ke papers ka free official PDF internet par downloadable nahi hai (mirrors sirf ~2019 tak ke papers rakhte hain). Ye version un hi papers ke public web renders se word-by-word English questions lekar banaya gaya hai, tai ki students unhe PDF me padh/print kar sakein. Hindi version aur exact layout ke liye original paper dekhein. Ye unofficial study material hai.